Coupled Harmonic Oscillators on a Ring

Problem 2. (Coupled harmonic oscillators) Consider a linear system of N particles coupled by springs, governed by the Hamiltonian

$$H = \frac{1}{2}\sum_{i=1}^{N} p_i^2 + \frac{\omega^2}{2}\sum_{i=1}^{N}(q_i - q_{i+1})^2, \qquad q_{N+1}q_1. \tag{1}$$

The Poisson brackets for the canonical variables are

$$\{q_k, q_l\} = 0, \qquad \{p_k, p_l\} = 0, \qquad \{p_k, q_l\} = -\delta_{k,l}. \tag{2}$$

Part 1. *Hamiltonian equations, matrix A, its eigenvalues and eigenvectors.*

The Hamiltonian equations read

$$\dot{q}_k = \{q_k, H\} = \frac{\partial H}{\partial p_k} = p_k, \qquad \dot{p}_k = \{p_k, H\} = -\frac{\partial H}{\partial q_k} = -\omega^2\big(2q_i - q_{i-1} - q_{i+1}\big). \tag{3}$$

In vector form these become

$$\dot{\boldsymbol{q}} = \boldsymbol{p}, \qquad \dot{\boldsymbol{p}} = -\omega^2 A\,\boldsymbol{q}, \tag{4}$$

where the matrix A encodes the nearest-neighbour coupling with periodic boundary conditions:

$$A = \begin{pmatrix} 2 & -1 & 0 & \cdots & \cdots & -1 \\ -1 & 2 & -1 & 0 & \cdots & 0 \\ 0 & -1 & 2 & -1 & \cdots & 0 \\ \vdots & \vdots & & \ddots & & \vdots \\ \vdots & \vdots & & & \ddots & \vdots \\ -1 & 0 & \cdots & \cdots & -1 & 2 \end{pmatrix}. \tag{5}$$

Eigenvalues and eigenvectors. The matrix A commutes with the cyclic permutation (translation) operator

$$T = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ \vdots & & & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix}, \tag{6}$$

so A and T share a common eigenbasis. Since $T^N = \mathbf{1}$, the eigenvalues of T satisfy $\lambda_k^N = 1$, giving

$$\lambda_k = e^{i2\pi k/N}, \qquad k = 0, 1, \dots, N-1. \tag{7}$$

The corresponding eigenvectors of T (and hence of A) are the discrete Fourier modes

$$\boldsymbol{x}_k = \big(1,\ e^{i2\pi k/N},\ \dots,\ e^{i2\pi k(N-1)/N}\big)^\top. \tag{8}$$

Applying A to $\boldsymbol{x}_k$, the j-th component gives

$$\begin{aligned}(A\boldsymbol{x}_k)_j &= e^{i2\pi kj/N}\big(2 - e^{i2\pi k/N} - e^{-i2\pi k/N}\big) \\ &= 2e^{i2\pi kj/N}(1 - \cos(2\pi k/N)) \\ &= 4\sin^2\!\Big(\frac{\pi k}{N}\Big)\,(\boldsymbol{x}_k)_j.\end{aligned} \tag{9}$$

Therefore the eigenvalues of A are

$$\mu_k = 4\sin^2\!\Big(\frac{\pi k}{N}\Big), \qquad k = 0, 1, \dots, N-1. \tag{10}$$

Note that $\mu_0 = 0$ (the zero mode corresponding to uniform translation) and $\mu_k = \mu_{N-k}$ (doubly degenerate for $k \neq 0, N/2$).

Part 2. *Normal coordinates, Hamiltonian, and solution.*

Set $N = 2M+1$ and define the discrete Fourier transforms

$$Q_k = \frac{1}{\sqrt{N}}\sum_{i=1}^{N} e^{\frac{2\pi i}{N}ki}\,q_i, \qquad P_k = \frac{1}{\sqrt{N}}\sum_{i=1}^{N} e^{\frac{2\pi i}{N}ki}\,p_i, \qquad k = -M, \dots, M. \tag{11}$$

The inverse transformation is

$$q_j = \frac{1}{\sqrt{N}}\sum_{k=-M}^{M} e^{\frac{-2\pi i}{N}kj}\,Q_k, \qquad p_j = \frac{1}{\sqrt{N}}\sum_{k=-M}^{M} e^{\frac{-2\pi i}{N}kj}\,P_k. \tag{12}$$

Hamiltonian in normal coordinates. Using the discrete orthogonality relation $\frac{1}{N}\sum_j e^{i2\pi(k+m)j/N} = \delta_{k,-m}$, we compute

$$\begin{aligned}\sum_j p_j^2 &= \frac{1}{N}\sum_{k,m,j} P_k P_m\, e^{\frac{i2\pi(k+m)j}{N}} = \sum_k P_{-k}P_k = \sum_k |P_k|^2, \\ \sum_{j=1}^{N}(q_j - q_{j+1})^2 &= \frac{1}{N}\sum_{k,m,j} Q_k Q_m\, e^{-i\pi(k+m)(2j+1)/N}\,\big(2i\sin\tfrac{\pi k}{N}\big)\big(2i\sin\tfrac{\pi m}{N}\big) \\ &= 4\sum_k |Q_k|^2 \sin^2\!\Big(\frac{\pi k}{N}\Big).\end{aligned} \tag{13}$$

Hence the Hamiltonian becomes a sum of decoupled oscillators:

$$\boxed{H = \frac{1}{2}\sum_{k=-M}^{M} |P_k|^2 + 2\omega^2 \sum_{k=-M}^{M} |Q_k|^2 \sin^2\!\Big(\frac{\pi k}{N}\Big).} \tag{14}$$

Equations of motion and their solution. The Hamiltonian equations in normal coordinates are

$$\dot{Q}_k = P_k, \qquad \dot{P}_k = -4\omega^2 \sin^2\!\Big(\frac{\pi k}{N}\Big) Q_k, \tag{15}$$

which combine into $\ddot{Q}_k = -\omega(k)^2 Q_k$ with dispersion relation

$$\omega(k) = 2\omega \sin\frac{\pi k}{N}. \tag{16}$$

The general solution is

$$Q_k(t) = Ae^{i\omega(k)t} + Be^{-i\omega(k)t}, \qquad P_k(t) = i\omega(k)\big(Ae^{i\omega(k)t} - Be^{-i\omega(k)t}\big). \tag{17}$$

The zero mode ($k = 0$) satisfies $\dot{P}_0 = 0$, so $Q_0(t) = Q_0(0) + P_0 t$ describes free centre-of-mass motion.

Part 3. *Creation/annihilation variables, conserved quantities, involutivity.*

For $k = \pm 1, \ldots, \pm M$, define

$$a_k = \frac{1}{\sqrt{2}}\big(P_{-k} + i\omega(k)Q_{-k}\big), \qquad \bar{a}_k = \frac{1}{\sqrt{2}}\big(P_k + i\omega(k)Q_k\big). \tag{18}$$

Hamiltonian in terms of a_k, $\bar{a}_k$. A direct computation gives

$$\begin{aligned} a_{-k}a_k &= \tfrac{1}{2}\big(|P_k|^2 + \omega(k)^2|Q_k|^2 + i\omega(k)(P_kQ_{-k} - P_{-k}Q_k)\big), \\ \bar{a}_{-k}\bar{a}_k &= \tfrac{1}{2}\big(|P_k|^2 + \omega(k)^2|Q_k|^2 + i\omega(k)(P_{-k}Q_k - P_kQ_{-k})\big), \end{aligned} \tag{19}$$

so that $a_{-k}a_k + \bar{a}_{-k}\bar{a}_k = |P_k|^2 + \omega(k)^2|Q_k|^2$. Since $2\omega^2\sin^2(\pi k/N) = \frac{1}{2}\omega(k)^2$, the Hamiltonian becomes

$$\boxed{H = \frac{1}{2}P_0^2 + \sum_{k=1}^{M}(a_{-k}a_k + \bar{a}_{-k}\bar{a}_k) = I_0 + \sum_{k=1}^{M}(I_k + I_{-k}).} \tag{20}$$

Poisson brackets. From $\{P_k, Q_l\} = -\frac{1}{N}\sum_j e^{i2\pi j(k+l)/N} = -\delta_{k,-l}/N$ we derive the fundamental brackets of the normal coordinates, and from those:

$$\begin{aligned} \{a_k, a_l\} &= \frac{i\omega(k)}{N}\,\delta_{k,-l}, \\ \{\bar{a}_k, \bar{a}_l\} &= \frac{i\omega(k)}{N}\,\delta_{k,-l}, \\ \{a_k, \bar{a}_l\} &= 0. \end{aligned} \tag{21}$$

The last bracket vanishes because the cross terms cancel:

$$2\{a_k, \bar{a}_l\} = i\omega(k)\{Q_{-k}, P_l\} + i\omega(l)\{P_{-k}, Q_l\} = \frac{i\omega(k)}{N}\delta_{k,l} - \frac{i\omega(l)}{N}\delta_{k,l} = 0. \tag{22}$$

Conservation laws. We show $\dot{I}_0 = \dot{I}_k = \dot{I}_{-k} = 0$ by verifying that I_0, I_k, I_{-k} each Poisson-commute with H.

Claim 1. *$\{I_k, I_l\} = 0$ for all $k \neq l$, where $I_k = a_{-k}a_k$ and $I_{-k} = \bar{a}_{-k}\bar{a}_k$.*

Proof. Using the Leibniz rule and the bracket relations computed above:

$$\begin{aligned}
\{a_{-k}a_k,\ a_{-l}a_l\} &= a_{-k}\{a_k, a_{-l}\}a_l + a_{-k}a_{-l}\{a_k, a_l\} + \{a_{-k}, a_{-l}\}a_k a_l + a_{-l}\{a_{-k}, a_l\}a_k \\
&= \frac{i\omega(k)a_{-k}a_k}{N}(\delta_{k,-l} + \delta_{k,l} - \delta_{k,l} - \delta_{k,-l}) = 0, \\
\{\bar{a}_{-k}\bar{a}_k,\ \bar{a}_{-l}\bar{a}_l\} &= 0 \quad \text{(same argument)}, \\
\{a_{-k}a_k,\ \bar{a}_{-l}\bar{a}_l\} &= 0 \quad (\text{since } \{a_\bullet, \bar{a}_\bullet\} = 0).
\end{aligned} \tag{23}$$

□

Since $H = I_0 + \sum_{k=1}^{M}(I_k + I_{-k})$ and $I_0 = \frac{1}{2}P_0^2$ trivially commutes with all I_k, we conclude

$$\{I_0, H\} = 0, \qquad \{I_k, H\} = \sum_{l=1}^{M}\{I_k, I_l + I_{-l}\} = 0, \qquad \{I_{-k}, H\} = 0, \tag{24}$$

hence $\dot{I}_0 = \dot{I}_k = \dot{I}_{-k} = 0$ for all $k = 1, \ldots, M$.

The system therefore possesses $2M+1 = N$ independent conserved quantities $\{I_0, I_1, \ldots, I_M, I_{-1}, \ldots, I_{-M}\}$ which are pairwise in involution, confirming complete integrability of the coupled oscillator chain.